Static Timing Analysis of VSDBABYSOC

SKY130 PVT Corner Assessment — Final Report

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Design	VSDBABYSOC (synthesized netlist with SDC constraints)
Tool	OpenSTA
Libraries	SKY130 sky130_fd_sc_hd Liberty PVT corners (PLL and DAC libraries unchanged)

Objective. Perform Static Timing Analysis (STA) of the VSDBABYSOC design using the provided SKY130 standard-cell Liberty corner libraries and observe timing behavior across process, voltage, and temperature (PVT) conditions.

Scope. 18 assessment Liberty corners were analyzed. The original TT 25 °C, 1.80 V library is retained as the baseline/reference (Idx 0). Two of the 18 corners are CCS-noise libraries; their values are kept in the results table but they are left out of the plots, which therefore show 16 valid corners.

18	**8 of 16**	**8 of 16**	**−62.56 ns**
assessment corners analyzed, plus the TT 25 °C baseline	valid corners report no negative slack (all FF and TT)	valid corners report negative slack (all SS)	worst WNS: SS, −40 °C, 1.28 V (TNS −13,957.60 ns)

1. PVT Corner Meaning

PVT stands for **Process, Voltage, and Temperature**. Each Liberty file models the standard cells at one such operating point, and the file name encodes it, for example `sky130_fd_sc_hd__ss_n40C_1v28.lib` = SS process, −40 °C, 1.28 V.

Field	Meaning	Values in this assessment
Process (P)	Process-speed characterization of the library	FF = Fast-Fast, TT = Typical-Typical, SS = Slow-Slow
Temperature	Temperature the library is characterized at	n40C = −40 °C, 025C = 25 °C, 100C = 100 °C
Voltage (V)	Supply voltage encoded in the file name suffix	1v28 = 1.28 V, 1v65 = 1.65 V, 1v80 = 1.80 V
ccsnoise	Variant of the library with CCS-noise tables	ff_n40C_1v95_ccsnoise and ss_n40C_1v60_ccsnoise

2. STA Methodology

Each Liberty corner was analyzed separately while the PLL and DAC libraries were kept unchanged. The same synthesized VSDBABYSOC netlist and SDC constraints were used for every run.

Detailed timing was reported per corner with:

```
report_checks -path_delay min_max -fields {nets cap slew input_pins fanout}
```

WNS / TNS were extracted with OpenSTA `report_wns` and `report_tns` for each corner, then collected from the per-corner report files in `lib/assessment/reports/` into summary files (Appendices A and B).

Reading the numbers. WNS (Worst Negative Slack) is the worst timing slack; TNS (Total Negative Slack) is the sum of negative slack over all violating endpoints. A value of 0.00 ns means OpenSTA reported no negative slack; it does not quantify positive margin.

3. WNS / TNS Results

Table 1 reproduces the results from the submitted Excel assessment workbook. Idx 0 is the baseline and the remaining 18 rows are the assessment corners. Failing corners are shown in red.

Idx	Library	P	Temp	V	WNS (ns)	TNS (ns)	Status
0	sky130_fd_sc_hd__tt_025C_1v80.lib	TT	25°C	1.80 V	0.00	0.00	Baseline
1	sky130_fd_sc_hd__ff_100C_1v65.lib	FF	100°C	1.65 V	0.00	0.00	Assessment
2	sky130_fd_sc_hd__ff_100C_1v95.lib	FF	100°C	1.95 V	0.00	0.00	Assessment
3	sky130_fd_sc_hd__ff_n40C_1v56.lib	FF	-40°C	1.56 V	0.00	0.00	Assessment
4	sky130_fd_sc_hd__ff_n40C_1v65.lib	FF	-40°C	1.65 V	0.00	0.00	Assessment
5	sky130_fd_sc_hd__ff_n40C_1v76.lib	FF	-40°C	1.76 V	0.00	0.00	Assessment
6	sky130_fd_sc_hd__ff_n40C_1v95.lib	FF	-40°C	1.95 V	0.00	0.00	Assessment
7	*sky130_fd_sc_hd__ff_n40C_1v95_ccsnoise.lib*	*FF*	*-40°C*	*1.95 V*	*0.00*	*0.00*	*Excluded — CCS-noise*
8	sky130_fd_sc_hd__ss_100C_1v40.lib	SS	100°C	1.40 V	**-17.12**	**-2599.51**	Assessment
9	sky130_fd_sc_hd__ss_100C_1v60.lib	SS	100°C	1.60 V	**-7.87**	**-892.70**	Assessment
10	sky130_fd_sc_hd__ss_n40C_1v28.lib	SS	-40°C	1.28 V	**-62.56**	**-13957.60**	Assessment
11	sky130_fd_sc_hd__ss_n40C_1v35.lib	SS	-40°C	1.35 V	**-39.70**	**-7993.79**	Assessment
12	sky130_fd_sc_hd__ss_n40C_1v40.lib	SS	-40°C	1.40 V	**-29.69**	**-5466.48**	Assessment
13	sky130_fd_sc_hd__ss_n40C_1v44.lib	SS	-40°C	1.44 V	**-23.91**	**-4107.19**	Assessment
14	sky130_fd_sc_hd__ss_n40C_1v60.lib	SS	-40°C	1.60 V	**-10.61**	**-1355.80**	Assessment
15	*sky130_fd_sc_hd__ss_n40C_1v60_ccsnoise.lib*	*SS*	*-40°C*	*1.60 V*	***-10.61***	***-1355.80***	*Excluded — CCS-noise*
16	sky130_fd_sc_hd__ss_n40C_1v76.lib	SS	-40°C	1.76 V	**-4.38**	**-367.93**	Assessment
17	sky130_fd_sc_hd__tt_025C_1v80.lib	TT	25°C	1.80 V	0.00	0.00	Assessment
18	sky130_fd_sc_hd__tt_100C_1v80.lib	TT	100°C	1.80 V	0.00	0.00	Assessment

Notes. Idx 0 and Idx 17 are the same library (sky130_fd_sc_hd__tt_025C_1v80.lib); the reports directory holds a single report for it. Idx 7 and 15 (shaded amber) are the CCS-noise corners: OpenSTA reported CCS-noise table-template warnings for them, so they are retained here but excluded from the plots.

4. Comparative Plots

The plots use the 16 valid assessment corners (CCS-noise corners excluded). Slack is plotted in its natural sign, so negative slack extends downward from the 0 ns line; corners with no negative slack appear as a flat marker on that line.

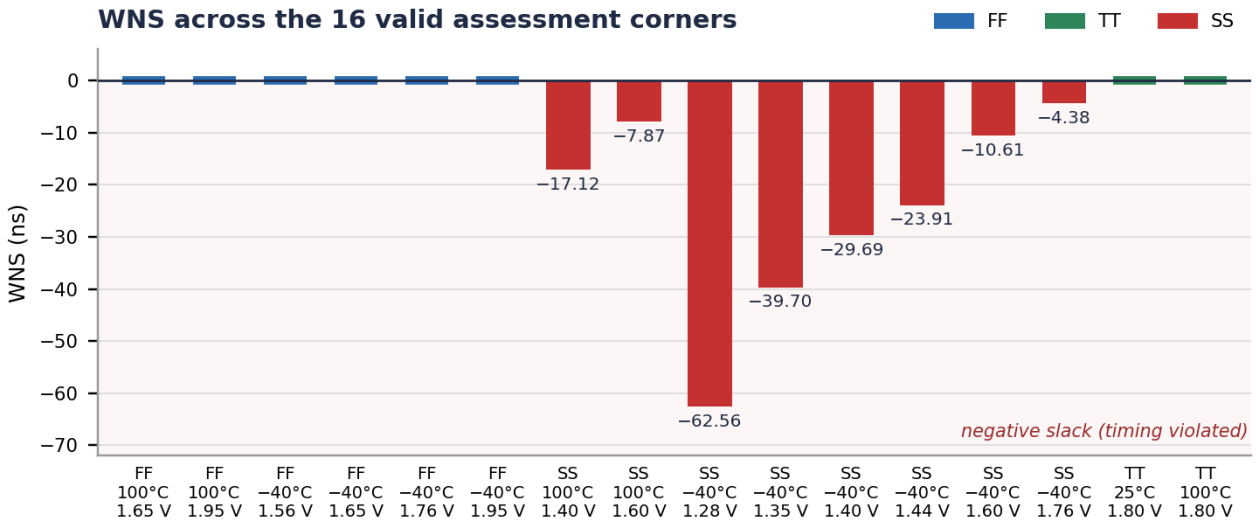


Figure 1. WNS across the 16 valid assessment corners.

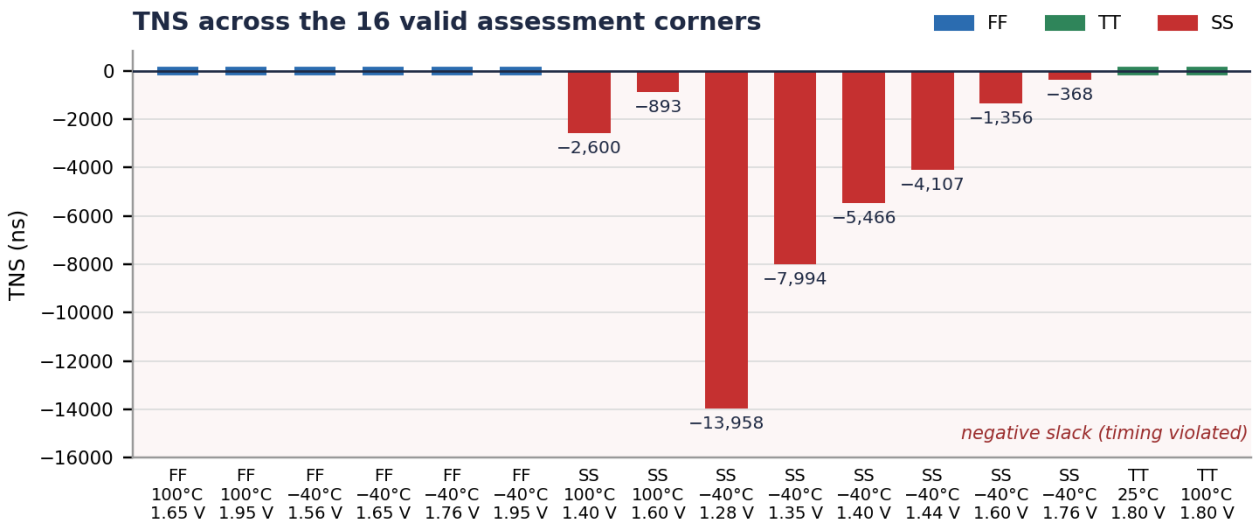


Figure 2. TNS across the 16 valid assessment corners (bar labels rounded to the nearest ns; exact values in Table 1).

5. Key Observations

1. **Only SS corners violate.** All six valid FF corners and both valid TT corners report 0.00 ns WNS and TNS. All eight valid SS corners report negative slack.
2. **Supply voltage dominates at SS, −40 °C.** WNS improves monotonically as the supply rises, from −62.56 ns at 1.28 V to −4.38 ns at 1.76 V (a 58.18 ns improvement over 0.48 V), and TNS improves from −13,957.60 ns to −367.93 ns. Even at 1.76 V this corner does not reach 0.00 ns (Figure 3).
3. **At SS, the cold corner is worse than the hot corner at the same supply.** At 1.40 V, WNS is −29.69 ns at −40 °C versus −17.12 ns at 100 °C; at 1.60 V it is −10.61 ns versus −7.87 ns. Slower behavior at low temperature when the supply is reduced is consistent with temperature inversion, although this assessment did not isolate the cause.
4. **Violations are widespread, not a few isolated paths.** Since |TNS| cannot exceed the number of violating endpoints times |WNS|, the ratio |TNS| / |WNS| gives a lower bound on that count: roughly 223 endpoints at SS, −40 °C, 1.28 V, falling to roughly 84 at 1.76 V (about 152 and 113 at SS, 100 °C, 1.40 V and 1.60 V).

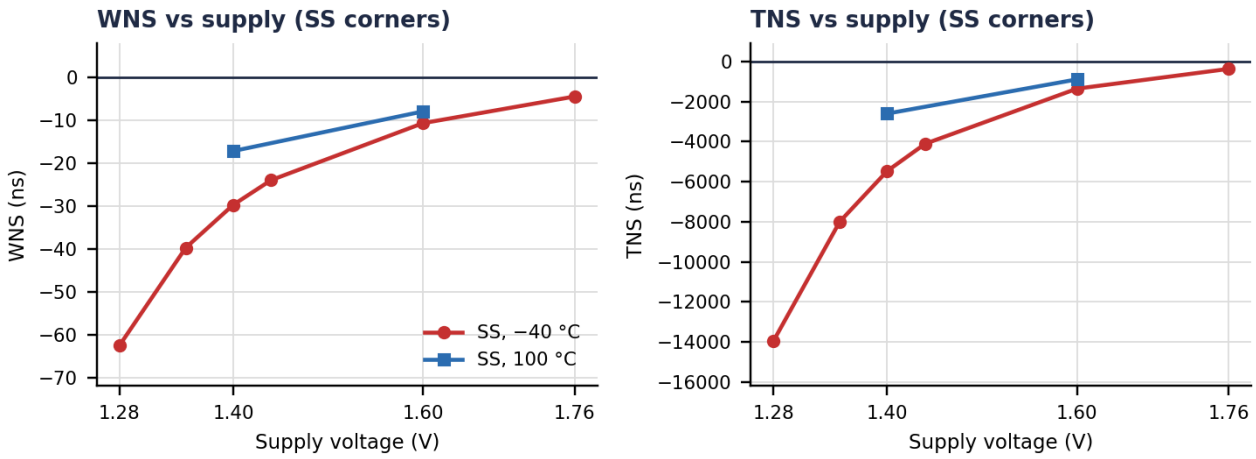


Figure 3. WNS and TNS versus supply voltage at the SS corners (−40 °C: 1.28–1.76 V; 100 °C: 1.40 and 1.60 V).

6. CCS-noise Corner Note

The assessment included two CCS-noise Liberty files: `sky130_fd_sc_hd__ff_n40C_1v95_ccsnoise.lib` and `sky130_fd_sc_hd__ss_n40C_1v60_ccsnoise.lib`. OpenSTA reported CCS-noise table-template warnings for these libraries. Their observed WNS/TNS values were retained in the results table, while both corners were excluded from the comparative plots to avoid presenting warning-affected corners alongside the standard timing corners without qualification.

The observed values match their non-CCS counterparts (FF, −40 °C, 1.95 V: 0.00 / 0.00 ns; SS, −40 °C, 1.60 V: −10.61 / −1355.80 ns), so excluding them does not change any conclusion.

7. Conclusion

STA was performed for all 18 provided assessment corners. FF and TT corners report 0.00 ns WNS/TNS, while SS corners exhibit negative slack to varying degrees, growing steadily as the supply voltage drops. The most negative observed WNS/TNS in the standard assessment set is at SS, −40 °C, 1.28 V: −62.56 ns WNS and −13,957.60 ns TNS.

The accompanying Excel workbook contains the complete numerical table and native Excel plots. This report presents the same results together with the compressed STA-output screenshots (Appendix A) and the raw WNS/TNS extraction (Appendix B).

Appendix A. STA Output Screenshots

Terminal captures of the compact summary files `compact_part1.txt` to `compact_part4.txt`, which together cover all 18 assessment corners.

```
vsduser@vsdsquadron:~/Desktop/work/tools/openlane_working_dir/OpenSTA/vsdbabysoc$ cat lib/assessment/reports/compact_part1.txt
VSDBABYSOC STA — COMPACT RESULTS (PART 1/4)
================================================

VSDBABYSOC — SKY130 PVT CORNER WNS/TNS SUMMARY
================================================

LIBRARY: sky130_fd_sc_hd__ff_100C_1v65.lib
PROCESS: FF
TEMPERATURE: 100°C
VOLTAGE: 1.65 V
WNS: 0.00 ns
TNS: 0.00 ns
STATUS: VALID ASSESSMENT CORNER

------------------------------------------------

LIBRARY: sky130_fd_sc_hd__ff_100C_1v95.lib
PROCESS: FF
TEMPERATURE: 100°C
VOLTAGE: 1.95 V
WNS: 0.00 ns
TNS: 0.00 ns
STATUS: VALID ASSESSMENT CORNER

------------------------------------------------

LIBRARY: sky130_fd_sc_hd__ff_n40C_1v56.lib
PROCESS: FF
TEMPERATURE: -40°C
VOLTAGE: 1.56 V
WNS: 0.00 ns
TNS: 0.00 ns
STATUS: VALID ASSESSMENT CORNER

------------------------------------------------

LIBRARY: sky130_fd_sc_hd__ff_n40C_1v65.lib
PROCESS: FF
TEMPERATURE: -40°C
VOLTAGE: 1.65 V
WNS: 0.00 ns
TNS: 0.00 ns
STATUS: VALID ASSESSMENT CORNER

------------------------------------------------

LIBRARY: sky130_fd_sc_hd__ff_n40C_1v76.lib
PROCESS: FF
TEMPERATURE: -40°C
VOLTAGE: 1.76 V
WNS: 0.00 ns
TNS: 0.00 ns
```

Figure 4. compact_part1.txt — Idx 1–5 (FF corners).

```
WNS: 0.00 ns
TNS: 0.00 ns
vsduser@vsdsquadron:~/Desktop/work/tools/openlane_working_dir/OpenSTA/vsdbabysoc$ cat lib/assessment/reports/compact_part2.txt
VSDBABYSOC STA — COMPACT RESULTS (PART 2/4)
================================================

LIBRARY: sky130_fd_sc_hd__ff_n40C_1v95_ccsnoise.lib
PROCESS: FF
TEMPERATURE: -40°C
VOLTAGE: 1.95 V
WNS: 0.00 ns
TNS: 0.00 ns
STATUS: CCS-NOISE WARNING / EXCLUDED FROM GRAPHS

------------------------------------------------

LIBRARY: sky130_fd_sc_hd__ff_n40C_1v95.lib
PROCESS: FF
TEMPERATURE: -40°C
VOLTAGE: 1.95 V
WNS: 0.00 ns
TNS: 0.00 ns
STATUS: VALID ASSESSMENT CORNER

------------------------------------------------

LIBRARY: sky130_fd_sc_hd__ss_100C_1v40.lib
PROCESS: SS
TEMPERATURE: 100°C
VOLTAGE: 1.40 V
WNS: -17.12 ns
TNS: -2599.51 ns
STATUS: VALID ASSESSMENT CORNER

------------------------------------------------

LIBRARY: sky130_fd_sc_hd__ss_100C_1v60.lib
PROCESS: SS
TEMPERATURE: 100°C
VOLTAGE: 1.60 V
WNS: -7.87 ns
TNS: -892.70 ns
STATUS: VALID ASSESSMENT CORNER

------------------------------------------------

LIBRARY: sky130_fd_sc_hd__ss_n40C_1v28.lib
PROCESS: SS
TEMPERATURE: -40°C
VOLTAGE: 1.28 V
WNS: -62.56 ns
TNS: -13957.60 ns
STATUS: VALID ASSESSMENT CORNERvsduser@vsdsquadron:~/Desktop/work/tools/openlane_working_dir/OpenSTA/vsdbabysoc$
```

Figure 5. compact_part2.txt — Idx 6–10 (includes CCS-noise corner Idx 7).

```
TNS: -13957.60 ns
STATUS: VALID ASSESSMENT CORNERvsduser@vsdsquadron:~/Desktop/work/tools/openlane_working_dir/OpenSTA/vsdbabysoc$ cat lib/assessment/reports/compact_part3.txt
VSDBABYSOC STA — COMPACT RESULTS (PART 3/4)
================================================

LIBRARY: sky130_fd_sc_hd__ss_n40C_1v35.lib
PROCESS: SS
TEMPERATURE: -40°C
VOLTAGE: 1.35 V
WNS: -39.70 ns
TNS: -7993.79 ns
STATUS: VALID ASSESSMENT CORNER

------------------------------------------------

LIBRARY: sky130_fd_sc_hd__ss_n40C_1v40.lib
PROCESS: SS
TEMPERATURE: -40°C
VOLTAGE: 1.40 V
WNS: -29.69 ns
TNS: -5466.48 ns
STATUS: VALID ASSESSMENT CORNER

------------------------------------------------

LIBRARY: sky130_fd_sc_hd__ss_n40C_1v44.lib
PROCESS: SS
TEMPERATURE: -40°C
VOLTAGE: 1.44 V
WNS: -23.91 ns
TNS: -4107.19 ns
STATUS: VALID ASSESSMENT CORNER

------------------------------------------------

LIBRARY: sky130_fd_sc_hd__ss_n40C_1v60_ccsnoise.lib
PROCESS: SS
TEMPERATURE: -40°C
VOLTAGE: 1.60 V
WNS: -10.61 ns
TNS: -1355.80 ns
STATUS: CCS-NOISE WARNING / EXCLUDED FROM GRAPHS

------------------------------------------------

LIBRARY: sky130_fd_sc_hd__ss_n40C_1v60.lib
PROCESS: SS
TEMPERATURE: -40°C
VOLTAGE: 1.60 V
WNS: -10.61 ns
TNS: -1355.80 ns
STATUS: VALID ASSESSMENT CORNERvsduser@vsdsquadron:~/Desktop/work/tools/openlane_working_dir/OpenSTA/vsdbabysoc$
```

Figure 6. compact_part3.txt — Idx 11–15 (includes CCS-noise corner Idx 15).

```
VOLTAGE: 1.60 V
WNS: -10.61 ns
TNS: -1355.80 ns
STATUS: VALID ASSESSMENT CORNERvsduser@vsdsquadron:~/Desktop/work/tools/openlane_working_dir/OpenSTA/vsdbabysoc$ cat lib/assessment/reports/compact_part4.txt
VSDBABYSOC STA — COMPACT RESULTS (PART 4/4)
================================================

LIBRARY: sky130_fd_sc_hd__ss_n40C_1v76.lib
PROCESS: SS
TEMPERATURE: -40°C
VOLTAGE: 1.76 V
WNS: -4.38 ns
TNS: -367.93 ns
STATUS: VALID ASSESSMENT CORNER

------------------------------------------------

LIBRARY: sky130_fd_sc_hd__tt_025C_1v80.lib
PROCESS: TT
TEMPERATURE: 25°C
VOLTAGE: 1.80 V
WNS: 0.00 ns
TNS: 0.00 ns
STATUS: VALID ASSESSMENT CORNER

------------------------------------------------

LIBRARY: sky130_fd_sc_hd__tt_100C_1v80.lib
PROCESS: TT
TEMPERATURE: 100°C
VOLTAGE: 1.80 V
WNS: 0.00 ns
TNS: 0.00 ns
STATUS: VALID ASSESSMENT CORNERvsduser@vsdsquadron:~/Desktop/work/tools/openlane_working_dir/OpenSTA/vsdbabysoc$
```

Figure 7. compact_part4.txt — Idx 16–18.

Appendix B. Raw WNS / TNS Extraction

Per-corner WNS and TNS values pulled directly from the individual OpenSTA report files.

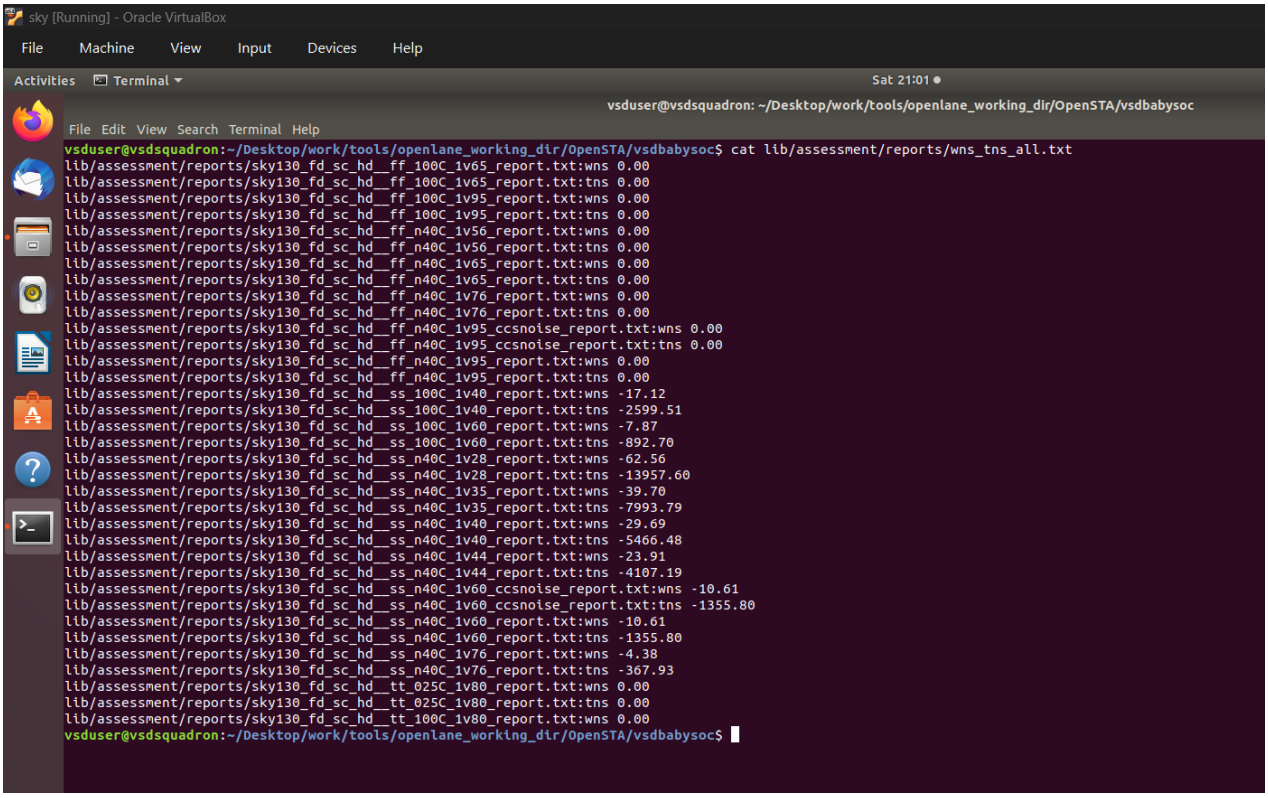


Figure 8. cat lib/assessment/reports/wns_tns_all.txt

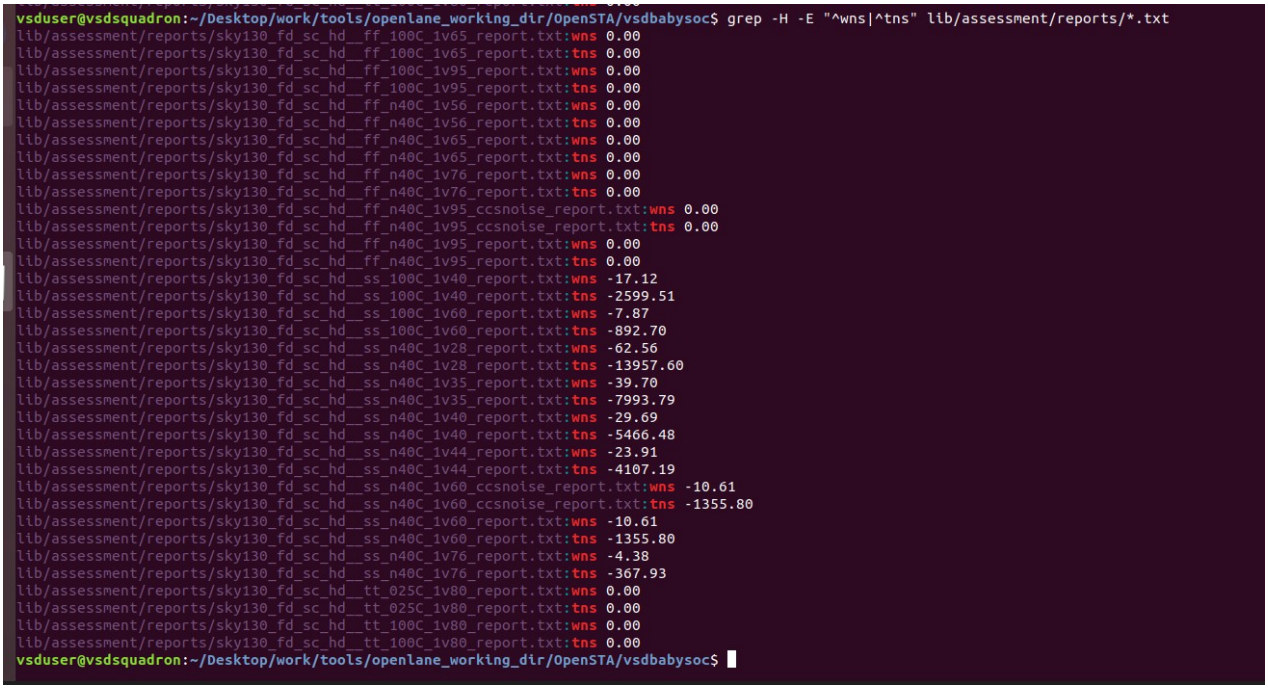


Figure 9. grep -H -E "^wns|^tns" lib/assessment/reports/.txt — WNS/TNS for every corner report.*